

Normal Modes

Chapter 8

So far, we have reviewed body waves and surface waves. The frequency band handled in seismology is in the range of $3 \times 10^{-4} \sim 100$ Hz. Oscillations with frequencies lower than approximately 0.005 Hz (5 mHz) are often treated as standing waves (Earth's free oscillations), whereas higher frequencies are typically treated as propagating waves (body waves, surface waves). In the Earth's free oscillation frequency band, the wavelength exceeds 1000 km, making it more convenient to understand them as superpositions of standing waves. The vibration pattern of a standing wave is also referred to as a normal mode.

§8.1 Standing wave of the Earth: Earth's free oscillations

When a huge earthquake occurs, seismic waves with periods around several hundred seconds travel around the Earth multiple times. In such cases, it is easier to understand them represented by a superposition of standing waves rather than as propagating waves as described in the previous section. First, let us look at observed records of earthquakes.

It is known that when a huge earthquake occurs, seismic waves circle the Earth many times. For example, during the recent 2010 Chilean earthquake, surface waves (seismic waves propagating along the Earth's surface) traveling around the Earth were actually recorded by seismometers. A very slow oscillation with a period of 200 seconds was recorded as it propagated. As seismic waves travel around the Earth's surface multiple times, their amplitude decreases, their shape deforms, and they disappear after about a day. Figure 8.1 shows the surface wave recorded by a global broadband seismic network for the earthquake that occurred in Chile on April 1, 2014 (the entire horizontal axis is about 6 hours). We can clearly see the surface wave passing around the world.

On December 26, 2004, a huge earthquake occurred off the coast of Sumatra. Its magnitude exceeded 9, making it the largest earthquake in the last 50 years. In this earthquake, seismometers captured surface waves circling the Earth 8 times. After a day, the seismic waves appeared to have

disappeared, but in reality, the Earth continued to vibrate for more than a month. This earthquake was so massive that the Earth actually kept vibrating for an even longer period.

Figure 8.2 shows the seismogram recorded at the Matsushiro station in Nagano Prefecture, Japan. At a glance, it looks similar to Figure 8.1, but compare the lengths of the horizontal axes. This time, two months of records are plotted. The Sumatra earthquake is recorded near the center. The second panel from the top (bandpass filtered at 0.1-1 mHz) shows that the Earth continued to oscillate even after a month. This recorded pulsing vibration of the entire Earth is called Earth's free oscillation.

Let's look at the actual data to see how it pulses. The lower panel of Figure 8.2 shows seismograms (vertical components) from Germany, the USA, Australia, and Japan about one month after the Sumatra earthquake. Normally, at observation stations around the world, seismic waves are recorded in the order they propagate from the source. What about in Figure 8.2? It shows that all observation stations are moving up and down almost simultaneously. These records indicate that the entire Earth uniformly pulses and repeats expansion and contraction at a period of 1112 seconds. The amplitude of the vibration is about 0.03 mm. At first glance, the amplitude seems small, but only a massive earthquake can cause the entire Earth to pulse. However, once it starts shaking, it has the property of continuing to shake for a long time. During the Sumatra earthquake, it was confirmed to have continued shaking for three months. In addition to the vibration pattern (called a mode) in which the entire Earth repeatedly expands and contracts, a mode in which the whole Earth squashes like a football is also known.

Earthquakes can be likened to striking all the keys of a piano at once. Each piano key corresponds to a vibration pattern (a mode). It is known that any vibration pattern can be reproduced by combinations of hitting many keys. (Although a piano has too few keys to reproduce every sound...). Just as we imagine the shape of an instrument by listening to its sound, seismologists ponder the interior of the Earth by observing its free oscillations with seismometers.

The fundamental reason they can be counted as modes is because the Earth has a finite size. The frequency spacing of the modes is deeply related to the size of the Earth. Mathematically speaking, it reduces to an eigenvalue problem of ordinary differential equations. In the 1D case, it corresponds to a Fourier series expansion, and the vibration pattern of the mode (eigenfunction) corresponds to a sin function.

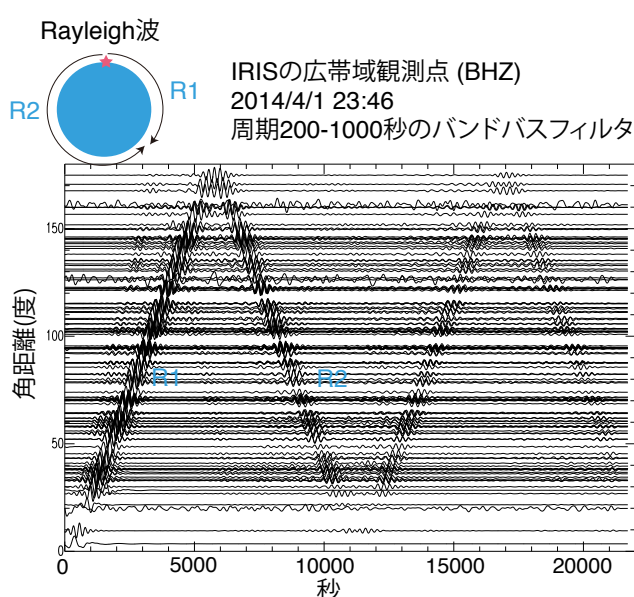


Figure 8.1: Vertical displacement records during the Chilean earthquake on April 1, 2014.

§8.2 Eigenvalue problem in 1D case: Oscillation of a string

Consider the oscillation of a string of length l . First, let us consider eigenfunctions that satisfy the wave equation:

$$\rho_0 \frac{\partial^2 u(x, t)}{\partial t^2} = \kappa_0 \frac{\partial^2 u(x, t)}{\partial x^2} \quad (8.1)$$

We assume fixed ends as boundary conditions at both ends. The eigenfrequencies and the corresponding eigenfunctions are given by:

$$\omega_n = \frac{n\pi}{l} c_0, \quad u_n(x) = A_n \sin \frac{n\pi x}{l} \quad (8.2)$$

For normalization, we determine the coefficient A_n to satisfy $(\rho_0 u_i, u_j) = \delta_{ij}$. Calculating the inner product,

$$(\rho_0 u_i, u_j) = \int_0^l \rho_0 u_i^*(x) u_j(x) dx = \rho_0 A_i^* A_j \int_0^l \sin \frac{i\pi x}{l} \sin \frac{j\pi x}{l} dx = \rho_0 |A_i|^2 \frac{l}{2} \delta_{ij} \quad (8.3)$$

Thus, we obtain

$$A_n = \sqrt{\frac{2}{\rho_0 l}} \quad \text{i.e.} \quad u_n(x) = \sqrt{\frac{2}{\rho_0 l}} \sin \frac{n\pi x}{l} \quad (8.4)$$

which is the normalized eigenfunction $u_n(x)$ ^{note 1)}.

Next, let us consider the response to an impulsive force:

$$\rho_0 \frac{\partial^2 u(x, t)}{\partial t^2} = \kappa_0 \frac{\partial^2 u(x, t)}{\partial x^2} + \delta(x - x_0) \delta(t) \quad (8.5)$$

This problem represents the motion of a rod fixed at both ends when an impulse of 1 [Ns] is applied ^{note 2)}. Taking the Fourier transform of both sides with respect to time yields:

$$-\omega^2 \rho_0 \tilde{u}(x, \omega) = \kappa_0 \frac{\partial^2 \tilde{u}(x, \omega)}{\partial x^2} + \delta(x - x_0), \quad (8.6)$$

where $\tilde{}$ denotes the Fourier component. Substituting this and the eigenfunction expansion of the wave solution $u(x, \omega)$

$$\tilde{u}(x, \omega) = \sum_{n=1}^{\infty} a_n(\omega) u_n(x) \quad (8.7)$$

^{note 1)} From the normalization of the eigenfunction, we can see that u_n^{-2} has dimensions of mass. This value is called the modal mass.

^{note 2)} Although the dimensions may seem confusing at a glance, $f(x) = -f_0 \delta(x - x_0) \delta(t)$ has dimensions of [N/m], so f_0 has dimensions of [Ns]. The vibration for an arbitrary force $f(x, t)$ can be represented by a convolution integral with the impulse response (called the Green's function) derived below. For an infinite space, see the Green's function section ??.

into equation (2), and taking the inner product of both sides with the eigenfunction $u_n(x)$, we obtain

$$\rho_0 \omega^2 a_n(\omega) - \rho_0 \omega_n^2 a_n(\omega) = -\rho_0 u_n(x_0) \quad \text{i.e.} \quad a_n(\omega) = -\frac{u_n(x_0)}{\omega^2 - \omega_n^2} \quad (8.8)$$

Finally, we return to the time domain by applying the inverse Fourier transform ^{note 3)},

$$a_n(t) = -\frac{1}{2\pi} \int_{-\infty}^{\infty} a_n(\omega) e^{-i\omega t} d\omega = -\frac{u_n(x_0)}{2\pi} \int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\omega^2 - \omega_n^2} d\omega \quad (8.9)$$

To evaluate this integral, we treat ω as a variable in the complex plane and integrate along the path shown in Figure 8.3, taking the limit $R \rightarrow \infty$. However, since the integral cannot be evaluated if there are poles on the integration path, we perform the integration for a function where the poles are shifted by $-i\epsilon$ ($\epsilon > 0$) from the real axis:

$$g(\omega, t, \epsilon) = \frac{e^{-i\omega t}}{2\omega_n} \left(\frac{1}{\omega - \omega_n + i\epsilon} - \frac{1}{\omega + \omega_n + i\epsilon} \right) \quad \left(\lim_{\epsilon \rightarrow 0} g(\omega, t, \epsilon) = \frac{e^{-i\omega t}}{\omega^2 - \omega_n^2} \right) \quad (8.10)$$

and then take the limit $\epsilon \rightarrow 0$. Here, there are four possible combinations to consider depending on whether the two poles are shifted in the positive or negative imaginary direction, but the solution satisfies the differential equation in any case. However, to satisfy the physical condition of causality (i.e., there is no displacement before the external force is applied at $t = 0$), the two poles must be shifted in the negative direction.

For $t > 0$, the poles are enclosed within the integration contour, so:

$$\begin{aligned} \int_{C1} g(\omega, t, \epsilon) d\omega + \int_{C2} g(\omega, t, \epsilon) d\omega \\ = -2\pi i \{ \text{Res}_{\omega=-i\epsilon+\omega_n} g(\omega, t, \epsilon) + \text{Res}_{\omega=-i\epsilon-\omega_n} g(\omega, t, \epsilon) \} = -\frac{2\pi \sin \omega_n t e^{-\epsilon t}}{\omega_n}. \end{aligned} \quad (8.11)$$

To prevent this expression from diverging as $t \rightarrow \infty$, the poles must be shifted in the negative direction. If they were shifted in the positive direction, it would be multiplied by an $e^{\epsilon t}$ term, which diverges as $t \rightarrow \infty$. Taking the limits $R \rightarrow \infty$ and $\epsilon \rightarrow 0$ in the above equation, the integral along path $C2$ becomes 0, yielding:

$$\int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\omega^2 - \omega_n^2} d\omega = -\frac{2\pi \sin \omega_n t}{\omega_n} \quad (t > 0). \quad (8.12)$$

On the other hand, for $t < 0$, there are no poles within the integration contour, so:

$$\int_{C1} g(\omega, t, \epsilon) d\omega + \int_{C3} g(\omega, t, \epsilon) d\omega = 0. \quad (8.13)$$

^{note 3)}For more details on complex integration, refer to Mathematical Methods for Physicists, Fifth Edition, Arfken, Weber, and Frank Harris,⁽¹⁾ or introductory textbooks on special functions and integral transforms.

Taking the limit $R \rightarrow \infty$ in the above equation, the integral along path $C3$ vanishes, yielding:

$$\int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\omega^2 - \omega_n^2} d\omega = 0 \quad (t < 0). \quad (8.14)$$

From the above, the desired $u(x, t)$ for $t > 0$ becomes, based on the string excitation equation:

$$u(x, t) = \begin{cases} \sum_n \frac{u_n^*(x_0)u_n(x)}{\omega_n} \sin \omega_n t & t \geq 0 \\ 0 & t < 0 \end{cases} \quad (8.15)$$

Substituting the normalized eigenfunctions into this equation, the required $u(x, t)$ for $t > 0$ is:

$$u(x, t) = \sum_{n=1}^{\infty} a_n(t)u_n(x) = \sum_{n=1}^{\infty} \left[\frac{2 \sin \omega_n t}{\rho_0 l \omega_n} \sin k_n x_0 \sin k_n x \right] \quad (8.16)$$

^{note 4)} Here we define $k_n = n\pi/l$.

To improve visibility, we rewrite the above equation using sum-to-product trigonometric identities ^{note 5)}.

$$u(x, t) = -\frac{1}{2\rho_0 l} \sum_{n=1}^{\infty} \frac{1}{\omega_n} [\sin(\omega_n t - k_n(x + x_0)) - \sin(\omega_n t + k_n(x + x_0)) \\ - \sin(\omega_n t - k_n(x - x_0)) + \sin(\omega_n t + k_n(x - x_0))] \quad (8.17)$$

Looking closely at $\sum_{n=1}^{\infty} \frac{\sin \omega_n t}{\omega_n}$, taking the derivative gives a delta function shape. This means it is related to the Heaviside step function $H(t)$ ^{note 6)}. It is a bit heuristic, but taking the Fourier series expansion of $H(x) - x/l$ in the interval $0 \leq x < l$, we get:

$$H(x) - x/l = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin k_n x, \quad (8.18)$$

Using this relation to rewrite the equation, we can express it as:

$$u(x, t) = \frac{1}{4\rho_0 c_0} \sum_{n=1}^{\infty} H_p(c_0 t + (x - x_0)) + H_p(c_0 t - (x - x_0)) - H_p(c_0 t - (x + x_0)) - H_p(c_0 t + (x + x_0)), \quad (8.19)$$

^{note 4)} Considering the dimensions of the equation provides a better perspective. When an impulse of 1 [Ns] is applied, we calculate the momentum change, divide by the modal mass to compute the velocity, and then infer the displacement by dividing by the frequency.

^{note 5)} For more details, refer to applied mathematics textbooks.

^{note 6)} The Heaviside step function $H(t)$ is defined as follows:

$$H(t) = \begin{cases} 1 & t > 0 \\ 0.5 & t = 0 \\ 0 & t < 0 \end{cases}$$

where $H_p(x) = H(x) - H(-x)$ for $-l < x < l$, assuming a periodicity of $2l$ in space. Figure 8.4 shows a schematic diagram of the propagation immediately after $t = 0$. At $t = 0$, the first two terms perfectly cancel each other out ^{note 7)}. Propagation begins from there, and immediately after at $t = t_0$, the displacement spreads out from the center. This can also be interpreted as the string trying to move upwards upon receiving the impulse. When the leftward propagating wave $H_p(c_0t + (x - x_0))$ reaches the left end $x = 0$, its phase reverses to satisfy the boundary condition and reflects (fixed end reflection). The reflection is represented by $H_p(c_0t - (x - x_0))$. Near $x = 0$, it cancels out the amplitude of $H_p(c_0t - (x - x_0))$, so the displacement becomes 0.

The results of summing up to $n = 1, 2, \dots, 40$ are shown in the left of Figure 8.5 and Figure 8.6. The parameters used are $\rho_0 = 1[\text{kg/m}]$, $\kappa_0 = 1[\text{N}]$, $l = 1[\text{m}]$, $x_0 = 0.3[\text{m}]$. In this case, information propagates at speed $c = 1[\text{m/s}]$. A positive amplitude always propagates left and right from the location where the force was applied. When reflected at the fixed end, as shown in Figure 8.4, the sign of one side is reversed. As a result, the reflected wave cancels out the amplitude of the other wave to become 0.

Also, the results when substituting $\sin(\omega_n t)/\omega_n$ with $\exp(-\omega_n^2 \tau_0^2) \cos(\omega_n t)$ and setting $\tau_0 = 0.01[\text{s}]$ are shown in the right of Figure 8.5 and Figure 8.6 ^{note 8)}. It can be observed that the pulse propagates in two directions from the source, and the phase reverses at both ends. Furthermore, the propagation velocity is $1[\text{m/s}]$, which is consistent with the velocity $c_0 = \sqrt{\kappa_0/\rho_0} = 1[\text{m/s}]$ calculated from the given parameters.

The following is supplementary information. Next, let us consider the momentum change of the string. Applying an impulse of 1 Ns means that the rod should carry a momentum of 1 Ns until it senses the fixed end. Therefore, we calculate the momentum of the rod. First, for simplicity, we assume $0 < x < l/2$ and consider the time interval $0 < t < x_0/c_0$. The momentum can be written as:

$$\rho_0 \frac{du(x, t)}{dt} = \frac{\delta(c_0t + (x - x_0)) + \delta(c_0t - (x - x_0))}{2}. \quad (8.20)$$

As mentioned in the footnote, the right graph of Figure 8.5 and the graph of Figure 8.6 correspond to velocity waveforms, which is consistent with this analytical solution for the velocity. Integrating this over the entire rod gives a value of $1[\text{Ns}]$. In the time interval $x_0/c_0 < t < (l - x_0)/c_0$,

$$\rho_0 \frac{du(x, t)}{dt} = \frac{-\delta(c_0t + (l - x_0)) + \delta(c_0t - (x - x_0))}{2}, \quad (8.21)$$

and integrating over the entire space gives a total momentum of $0[\text{Ns}]$. That is, when the wave reaches the left end, because it is a fixed end, it receives a downward impulse of $1[\text{Ns}]$ (the change in momentum).

^{note 7)}Note that we are considering $H_p(c_0t + (x - x_0))$ and $H_p(c_0t - (x - x_0))$ propagating in opposite directions. This is in the form of d'Alembert's solution and is intuitively easy to understand.

^{note 8)}This substitution can be interpreted as taking the time derivative of $\sin(\omega_n t)/\omega_n$ to get a velocity waveform, and applying a low-pass filter with a period of τ_0 .

8.2.1 Key Points

1. Do not forget the normalization. When κ_0, ρ_0 have spatial distributions, the eigenfunctions will not be orthogonal unless the inner product is defined including the density.
2. When performing complex integration, note that the integration path must be changed depending on the sign of t . From the above discussion, we see that causality is satisfied only when poles exist below the real axis. Note that if a principal value integral is considered, the contribution of the integral around the singularities does not vanish even after taking limits^{note 9)}.

§8.3 Eigenvalue Problem: The SH Case

First, let us organize the equation of motion (Equation 5.19) and Hooke's law (Equation 5.22) for the plane wave of the SH problem. Here, we take s_y, T_{xz} as variables to consider the boundary conditions. Since the governing equations consist only of partial derivatives with respect to z , the first-order ordinary differential equation with respect to z can be written as:

$$\frac{d}{dz} \begin{pmatrix} s_y \\ T_{xz} \end{pmatrix} = \begin{pmatrix} 0 & \frac{1}{\mu} \\ \omega^2 p^2 \mu - \rho \omega^2 & 0 \end{pmatrix} \begin{pmatrix} s_y \\ T_{xz} \end{pmatrix} \quad (8.22)$$

Also, T_{yx} can be computed from s_y as

$$T_{yx} = i\omega p \mu s_y. \quad (8.23)$$

We can think of this as integrating the ordinary differential equations from one boundary to find the parameter p that satisfies the required boundary conditions at the other end. This is nothing but an eigenvalue problem.

§8.4 Eigenvalue Problem for a Single-layer Structure

For simplicity, let us consider a case where the structure is homogeneous in the vertical z direction and consider it more concretely. To improve visibility, we rewrite the governing equations as:

$$\frac{d}{dz} \mathbf{y} = \mathbf{A} \mathbf{y}, \quad (8.24)$$

^{note 9)} Instead of complex integration, one can also laboriously solve the problem by considering the continuation of the solution at $t = 0$.

where,

$$\mathbf{A} \equiv \begin{pmatrix} 0 & \frac{1}{\mu} \\ \omega^2 p^2 \mu - \rho \omega^2 & 0 \end{pmatrix}, \quad (8.25)$$

$$\mathbf{y} \equiv \begin{pmatrix} s_y \\ T_{xz} \end{pmatrix}. \quad (8.26)$$

Here, we consider the eigenvalue problem for matrix $\mathbf{A}$. Let the i -th eigenvalue be η_i and the corresponding eigenvector be $\boldsymbol{\phi}_i$. From the characteristic equation $|\mathbf{A} - \eta_i \mathbf{I}| = 0$, we obtain

$$\eta_1 = \omega \sqrt{p^2 - \frac{1}{\beta^2}} \quad (8.27)$$

$$\eta_2 = -\omega \sqrt{p^2 - \frac{1}{\beta^2}} \quad (8.28)$$

The corresponding eigenvectors are

$$\boldsymbol{\phi}_1 = \begin{pmatrix} 1/\mu \\ \eta_1 \end{pmatrix} \quad (8.29)$$

$$\boldsymbol{\phi}_2 = \begin{pmatrix} 1/\mu \\ \eta_2 \end{pmatrix}. \quad (8.30)$$

They satisfy the relation

$$\mathbf{A} \boldsymbol{\phi}_i = \eta_i \boldsymbol{\phi}_i. \quad (8.31)$$

In this case, the matrix can be decomposed as

$$\mathbf{A} = (\boldsymbol{\phi}_1, \boldsymbol{\phi}_2) \begin{pmatrix} \eta_1 & \\ & \eta_2 \end{pmatrix} (\boldsymbol{\phi}_1, \boldsymbol{\phi}_2)^{-1} \equiv \boldsymbol{\Phi} \mathbf{H} \boldsymbol{\Phi}^{-1}, \quad (8.32)$$

where $\mathbf{H}$ is a diagonal matrix containing the eigenvalues η_i . By defining $\tilde{\mathbf{y}} \equiv \boldsymbol{\Phi}^{-1} \mathbf{y}$, we get

$$\frac{d}{dz} \tilde{\mathbf{y}} = \mathbf{H} \tilde{\mathbf{y}}, \quad (8.33)$$

which allows us to solve the differential equation component by component. Summarizing these solutions, the general solution can be written as

$$\mathbf{y} = \sum_{i=1}^2 A_i e^{\eta_i z} \boldsymbol{\phi}_i. \quad (8.34)$$

§8.5 Eigenvalue Problem: The P-SV Case

Here, let us organize the equation of motion (Equation 5.19) and Hooke's law (Equation 5.22) for the plane wave of the P-SV problem. We take s_x, s_z, T_{xz}, T_{zz} as variables to account for the boundary conditions. Since the governing equations consist only of partial derivatives with respect to z , the first-order ordinary differential equation in z can be written as:

$$\frac{d}{dz} \begin{pmatrix} s_x \\ s_z \\ T_{xz} \\ T_{zz} \end{pmatrix} = \begin{pmatrix} 0 & -i\omega p & \frac{1}{\mu} & 0 \\ -i\omega p \frac{\lambda}{\lambda+2\mu} & 0 & 0 & \frac{1}{\lambda+2\mu} \\ -\rho\omega^2 + \omega^2 p^2 \frac{4\mu(\lambda+\mu)}{\lambda+2\mu} & 0 & 0 & -i\omega p \frac{\lambda}{\lambda+2\mu} \\ 0 & -\rho\omega^2 & -i\omega p & 0 \end{pmatrix} \begin{pmatrix} s_x \\ s_z \\ T_{xz} \\ T_{zz} \end{pmatrix} \quad (8.35)$$

Here, T_{xx} can be determined from s_x and T_{zx} as

$$T_{xx} = i\omega p(\lambda + 2\mu)s_x + \lambda \frac{ds_z}{dz}. \quad (8.36)$$

This means the problem is restated as finding the parameter p by integrating the ordinary differential equations from one boundary such that they satisfy the given conditions at the other boundary. This is simply an eigenvalue problem. The problem configuration in the first part of section 5.6.2 corresponds to integrating from the bottom and matching the boundary condition at the surface, while the second part corresponds to starting from the surface and matching the solution at infinity as the boundary condition.

8.5.1 Eigenvalue Problem for Homogeneous Structure

For simplicity, we consider a homogeneous structure in the vertical z direction to examine this more concretely. To improve clarity, we rewrite the governing equations as:

$$\frac{d}{dz} \mathbf{y} = \mathbf{A} \mathbf{y}, \quad (8.37)$$

where,

$$\mathbf{A} \equiv \begin{pmatrix} 0 & -i\omega p & \frac{1}{\mu} & 0 \\ -i\omega p \frac{\lambda}{\lambda+2\mu} & 0 & 0 & \frac{1}{\lambda+2\mu} \\ -\rho\omega^2 + \omega^2 p^2 \frac{4\mu(\lambda+\mu)}{\lambda+2\mu} & 0 & 0 & -i\omega p \frac{\lambda}{\lambda+2\mu} \\ 0 & -\rho\omega^2 & -i\omega p & 0 \end{pmatrix}, \quad (8.38)$$

$$\mathbf{y} \equiv \begin{pmatrix} s_x \\ s_z \\ T_{xz} \\ T_{zz} \end{pmatrix}. \quad (8.39)$$

We now consider the eigenvalue problem of the matrix A . Assuming the i -th eigenvalue is η_i and the eigenvector is ϕ^i , the following relationship holds:

$$A\phi^i = \eta_i\phi^i. \quad (8.40)$$

It can thus be decomposed as

$$A = (\phi_1, \phi^2, \phi^3, \phi^4) \begin{pmatrix} \eta_1 & & & \\ & \eta_2 & & \\ & & \eta_3 & \\ & & & \eta_4 \end{pmatrix} (\phi_1, \phi^2, \phi^3, \phi^4)^{-1} \equiv \Phi H \Phi^{-1}, \quad (8.41)$$

where H is a diagonal matrix containing η_i . Defining $\tilde{y} \equiv \Phi^{-1}y$, we obtain

$$\frac{d}{dz}\tilde{y} = H\tilde{y}, \quad (8.42)$$

allowing the differential equation for each component to be explicitly solved. The general solution is written as a superposition:

$$y = \sum_{i=1}^4 A^i e^{\eta_i z} \phi^i. \quad (8.43)$$

8.5.2 Eigenvalues

Let us explicitly derive the eigenvalues. Calculating the determinant of the matrix $A - \eta I$ gives [note 10\)](#)

$$|A| = \left(\eta^2 - \omega^2 \left(p^2 - \frac{1}{\alpha^2} \right) \right) \left(\eta^2 - \omega^2 \left(p^2 - \frac{1}{\beta^2} \right) \right) = 0, \quad (8.45)$$

which factors into this equation. We observe that these eigenvalues correspond to the vertical propagation of seismic waves. Additionally, η_i represents an evanescent wave when η_i^2 is positive, and a body wave when it is negative.

8.5.3 Eigenvectors

[note 10\)](#) By straightforward cofactor expansion, one computes:

$$\begin{vmatrix} -\eta & a & b & 0 \\ c & -\eta & 0 & d \\ e & 0 & -\eta & c \\ 0 & f & a & -\eta \end{vmatrix} = \eta^4 - \eta^2(df + 2ac + be) + (ed - c^2)(bf - a^2) \quad (8.44)$$

Let us write down the corresponding eigenvectors. For simplicity, we assume the horizontal displacement is finite and normalize it such that $\phi_{i,1} = 1$:

$$\mathbf{A} \begin{pmatrix} \phi_{i,1} \\ \phi_{i,2} \\ \phi_{i,3} \\ \phi_{i,4} \end{pmatrix} = \eta_i \begin{pmatrix} \phi_{i,1} \\ \phi_{i,2} \\ \phi_{i,3} \\ \phi_{i,4} \end{pmatrix}. \quad (8.46)$$

We evaluate row by row, reducing the variables to $\phi_{i,3}$ and $\phi_{i,4}$ to calculate the eigenvectors.

Row 1

From the first row,

$$\phi_{i,2} = -\frac{i\phi_{i,3}}{\omega p \mu} + \frac{i\eta_i}{\omega p} \quad (8.47)$$

Row 2

Organizing the second row's components,

$$\eta_i \phi_{i,2} - \frac{1}{\lambda + 2\mu} \phi_{i,4} = -i\omega p \frac{\lambda}{\lambda + 2\mu} \quad (8.48)$$

Substituting Equation 8.47, we get

$$\frac{\eta_i}{\mu} \phi_{i,3} - \frac{i\omega p}{\lambda + 2\mu} \phi_{i,4} = \frac{\omega^2 p^2 \lambda}{\lambda + 2\mu} + \eta_i^2 \quad (8.49)$$

Row 4

$$-\rho\omega^2 \phi_{i,2} - i\omega p \phi_{i,3} - \eta_i \phi_{i,4} = 0 \quad (8.50)$$

Substituting Equation 8.47, we get

$$-\rho\omega^2 \left(-\frac{i\phi_{i,3}}{\omega p \mu} + \frac{i\eta_i}{\omega p} \right) - i\omega p \phi_{i,3} - \eta_i \phi_{i,4} = \quad (8.51)$$

$$\frac{\rho\omega i}{p\mu} \phi_{i,3} - i\omega p \phi_{i,3} - \eta_i \phi_{i,4} - \frac{i\rho\omega\eta_i}{p} = 0 \quad (8.52)$$

$$\left(\frac{i\rho\omega}{\mu p} - i\omega p \right) \phi_{i,3} - \eta_i \phi_{i,4} = \frac{i\rho\omega\eta_i}{p} \quad (8.53)$$

2-Component Eigenvalue Problem

Combining the equations for $\phi_{i,3}$ and $\phi_{i,4}$, we obtain

$$\begin{pmatrix} \eta_i & -\frac{i\omega p\mu}{\lambda+2\mu} \\ \frac{i\omega}{p} \left(p^2 - \frac{\rho}{\mu}\right) & \eta_i \end{pmatrix} \begin{pmatrix} \phi_{i,3} \\ \phi_{i,4} \end{pmatrix} = \begin{pmatrix} \frac{\omega^2 p^2 \lambda \mu}{\lambda+2\mu} + \eta_i^2 \mu \\ -\frac{i\rho\omega\eta_i}{p} \end{pmatrix} \quad (8.54)$$

By solving this system, $\phi_{i,3}$ and $\phi_{i,4}$ are explicitly expressed as:

$$\begin{pmatrix} \phi_{i,3} \\ \phi_{i,4} \end{pmatrix} = \begin{pmatrix} \eta_i & -\frac{i\omega p\mu}{\lambda+2\mu} \\ \frac{i\omega}{p} \left(p^2 - \frac{\rho}{\mu}\right) & \eta_i \end{pmatrix}^{-1} \begin{pmatrix} \frac{\omega^2 p^2 \lambda \mu}{\lambda+2\mu} + \eta_i^2 \mu \\ -\frac{i\rho\omega\eta_i}{p} \end{pmatrix} \quad (8.55)$$

$$= \frac{1}{\Delta_i} \begin{pmatrix} \eta_i & \frac{i\omega p\mu}{\lambda+2\mu} \\ -\frac{i\omega}{p} \left(p^2 - \frac{\rho}{\mu}\right) & \eta_i \end{pmatrix} \begin{pmatrix} \frac{\omega^2 p^2 \lambda \mu}{\lambda+2\mu} + \eta_i^2 \mu \\ -\frac{i\rho\omega\eta_i}{p} \end{pmatrix}, \quad (8.56)$$

$$= \frac{1}{\Delta_i} \begin{pmatrix} \eta_i & \frac{i\omega p\beta^2}{\alpha^2} \\ -\frac{i\omega}{p} \left(p^2 - \frac{1}{\beta^2}\right) & \eta_i \end{pmatrix} \begin{pmatrix} \rho\beta^2 \left(\frac{\omega^2 p^2 (\alpha^2 - 2\beta^2)}{\alpha^2} + \eta_i^2\right) \\ -\frac{i\rho\omega\eta_i}{p} \end{pmatrix} \quad (8.57)$$

$$= \frac{1}{\Delta_i} \begin{pmatrix} \eta_i \rho \beta^2 \left(\frac{\omega^2 p^2 (\alpha^2 - 2\beta^2)}{\alpha^2} + \eta_i^2\right) + \frac{\omega p \beta^2 \rho \omega \eta_i}{\alpha^2} \\ -\frac{i\omega}{p} \left(p^2 - \frac{1}{\beta^2}\right) \rho \beta^2 \left(\frac{\omega^2 p^2 (\alpha^2 - 2\beta^2)}{\alpha^2} + \eta_i^2\right) - \frac{i\rho\omega\eta_i^2}{p} \end{pmatrix} \quad (8.58)$$

$$= \frac{1}{\Delta_i} \begin{pmatrix} \eta_i \rho \beta^2 \left(\left(\omega^2 p^2 \left(1 - 2\frac{\beta^2}{\alpha^2}\right) + \eta_i^2\right) + \frac{\omega^2}{\alpha^2}\right) \\ -\frac{i\omega\rho}{p} \left(\beta^2 \left(p^2 - \frac{1}{\beta^2}\right) \left(\omega^2 p^2 \left(1 - 2\frac{\beta^2}{\alpha^2}\right) + \eta_i^2\right) + \eta_i^2\right) \end{pmatrix} \quad (8.59)$$

where Δ_i is the determinant of A ,

$$\Delta_i = \eta_i^2 - \frac{\omega^2 \mu}{\lambda + 2\mu} \left(p^2 - \frac{\rho}{\mu}\right) \quad (8.60)$$

$$= \begin{cases} \omega^2 p^2 \left(1 - \frac{\beta^2}{\alpha^2}\right), & i = 1, 2 \\ \omega^2 \left(p^2 - \frac{1}{\beta^2}\right) \left(1 - \frac{\beta^2}{\alpha^2}\right), & i = 3, 4 \end{cases} \quad (8.61)$$

Here, let us write out the corresponding eigenvectors case by case for each eigenvalue:

$$(\eta_1, \eta_2, \eta_3, \eta_4) = \omega \left(\sqrt{p^2 - \alpha^{-2}}, -\sqrt{p^2 - \alpha^{-2}}, \sqrt{p^2 - \beta^{-2}}, -\sqrt{p^2 - \beta^{-2}} \right) \quad (8.62)$$

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

For $i = 1, 2$, which correspond to P-wave propagation,

$$\begin{pmatrix} \phi_{i,3} \\ \phi_{i,4} \end{pmatrix} = \frac{1}{\Delta_i} \begin{pmatrix} \eta_i \rho \beta^2 \left(\left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \eta_i^2 \right) + \frac{\omega^2}{\alpha^2} \right) \\ -\frac{i\omega\rho}{p} \left(\beta^2 \left(p^2 - \frac{1}{\beta^2} \right) \left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \eta_i^2 \right) + \eta_i^2 \right) \end{pmatrix} \quad (8.63)$$

$$= \frac{1}{\Delta_i} \begin{pmatrix} \eta_i \rho \beta^2 \left(\left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \omega^2 (p^2 - \alpha^{-2}) \right) + \frac{\omega^2}{\alpha^2} \right) \\ -\frac{i\omega\rho}{p} \left(\beta^2 \left(p^2 - \frac{1}{\beta^2} \right) \left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \omega^2 (p^2 - \alpha^{-2}) \right) + \omega^2 (p^2 - \alpha^{-2}) \right) \end{pmatrix} \quad (8.64)$$

$$= \left(1 - \frac{\beta^2}{\alpha^2} \right) \frac{\omega^2 p \mu}{\Delta_i} \begin{pmatrix} 2\eta_i p \\ -i\omega \left(2p^2 - \frac{1}{\beta^2} \right) \end{pmatrix} \quad (8.65)$$

$$= \frac{\mu}{p} \begin{pmatrix} 2\eta_i p \\ -i\omega \left(2p^2 - \frac{1}{\beta^2} \right) \end{pmatrix} \quad (8.66)$$

$$\begin{pmatrix} \phi_{i,1} \\ \phi_{i,2} \\ \phi_{i,3} \\ \phi_{i,4} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{i\eta_i}{\omega p} \\ 2\eta_i \mu \\ -\frac{i\omega\mu}{p} \left(2p^2 - \frac{1}{\beta^2} \right) \end{pmatrix} = \frac{\omega}{p} \begin{pmatrix} \frac{p}{\omega} \\ -\frac{i\eta_i}{\omega^2} \\ 2\frac{\eta_i \mu p}{\omega} \\ -i\mu \left(2p^2 - \frac{1}{\beta^2} \right) \end{pmatrix} \quad (8.67)$$

On the other hand, for $i = 3, 4$, which correspond to S-wave propagation,

$$\begin{pmatrix} \phi_{i,3} \\ \phi_{i,4} \end{pmatrix} = \frac{1}{\Delta_i} \begin{pmatrix} \eta_i \rho \beta^2 \left(\left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \eta_i^2 \right) + \frac{\omega^2}{\alpha^2} \right) \\ -\frac{i\omega\rho}{p} \left(\beta^2 \left(p^2 - \frac{1}{\beta^2} \right) \left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \eta_i^2 \right) + \eta_i^2 \right) \end{pmatrix} \quad (8.68)$$

$$= \frac{1}{\Delta_i} \begin{pmatrix} \eta_i \rho \beta^2 \left(\left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \omega^2 (p^2 - \beta^{-2}) \right) + \frac{\omega^2}{\alpha^2} \right) \\ -\frac{i\omega\rho}{p} \left(\beta^2 \left(p^2 - \frac{1}{\beta^2} \right) \left(\omega^2 p^2 \left(1 - 2 \frac{\beta^2}{\alpha^2} \right) + \omega^2 (p^2 - \beta^{-2}) \right) + \omega^2 (p^2 - \beta^{-2}) \right) \end{pmatrix} \quad (8.69)$$

$$= \left(1 - \frac{\beta^2}{\alpha^2} \right) \frac{\omega^2 \mu}{\Delta_i} \begin{pmatrix} \eta_i \left(2p^2 - \frac{1}{\beta^2} \right) \\ -2i\omega p \left(p^2 - \frac{1}{\beta^2} \right) \end{pmatrix} \quad (8.70)$$

$$= \frac{\mu}{p^2 - \frac{1}{\beta^2}} \begin{pmatrix} \eta_i \left(2p^2 - \frac{1}{\beta^2} \right) \\ -2i\omega p \left(p^2 - \frac{1}{\beta^2} \right) \end{pmatrix} \quad (8.71)$$

$$\begin{pmatrix} \phi_{i,1} \\ \phi_{i,2} \\ \phi_{i,3} \\ \phi_{i,4} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{i\eta_i}{\omega} \frac{p}{p^2 - \frac{1}{\beta^2}} \\ 2p^2 - \frac{1}{\beta^2} \\ \mu\eta_i \frac{p^2 - \frac{1}{\beta^2}}{p^2 - \frac{1}{\beta^2}} \\ -2i\omega p \mu \end{pmatrix} = \frac{1}{\eta_i} \begin{pmatrix} \eta_i \\ -\frac{i p}{\omega} \\ \mu \left(2p^2 - \frac{1}{\beta^2} \right) \\ -2i\omega p \mu \eta_i \end{pmatrix} \quad (8.72)$$

The general solution can be formulated as a linear combination of the eigenvectors scaled by the phase contribution of vertical propagation e^{niz} .

8.5.4 Boundary Conditions

In Section 5.6, we analyzed Rayleigh waves propagating in a semi-infinite medium. Here, let us consider a free boundary surface and verify if we obtain identical results.

Assuming a z -axis with upward chosen as positive, let $z = 0$ be the free surface. First, because we are representing a semi-infinite medium, y_2 and y_4 are physically inadmissible as their amplitudes diverge at $z = -\infty$. Hence, the solution must be represented by a linear combination of y_1 and y_3 alone. At the free boundary condition $z = 0$, the stresses T_{xz} and T_{zz} must be zeroes, yielding:

$$\begin{vmatrix} \phi_{1,3} & \phi_{3,3} \\ \phi_{1,4} & \phi_{3,4} \end{vmatrix} = 0. \quad (8.73)$$

By writing out and simplifying this characteristic equation, we acquire

$$16 \left(-\frac{1}{\alpha^2} + \frac{1}{\beta^2} \right) p^6 + \left(\frac{16}{\alpha^2 \beta^2} - \frac{24}{\beta^4} \right) p^4 + \frac{8}{\beta^6} p^2 - \frac{1}{\beta^8} = 0. \quad (8.74)$$

We assert that this is identical to the characteristic equation of Rayleigh waves, from which we can ascertain the Rayleigh wave phase velocity.

Problem 8.1

1. Derive equation 8.74.

§8.6 Eigenvalue problem for spherically symmetric structure

When considering global-scale oscillations, it is a good approximation to treat the Earth as having structure only varying in the vertical direction, and homogeneous in the horizontal direction (spherically symmetric structure). In this case, the P-SV and SH problems correspond to spheroidal mode and toroidal mode oscillations, respectively. In this section, let us consider the oscillation of a homogeneous sphere, which is the simplest among spherically symmetric structures. For oscillations in more general spherically symmetric structures, refer to Saito's textbook⁽¹⁾ or Dahlen and Tromp (1998).⁽³⁾

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

§8.7 Eigenvibration and eigenfrequency

(4), (6), (10), (11), (13)

§8.8 Normal mode solutions for a homogeneous gas sphere

For simplicity, let us consider the eigenfrequencies and eigenfunctions for a homogeneous gas sphere. Here we assume that the pressure $P(\mathbf{x}; \omega)$ satisfies the wave equation

$$\nabla^2 P = -\frac{\omega^2}{c^2} P \quad (8.75)$$

The Laplacian in spherical coordinates can be written as

$$\nabla^2 = \frac{1}{r^2} \left(\frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} + \nabla_h^2 \right) \quad (8.76)$$

$$\nabla_h^2 = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \quad (8.77)$$

We assume that the pressure $P = R(r)Y(\theta, \varphi)$ can be separated by variables in the r , θ , φ directions into $R(r)$ and $Y(\theta, \varphi)$.

$$\frac{1}{r^2} \left[\frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{\nabla_h^2 Y}{Y} \right] = -\frac{\omega^2}{c^2}. \quad (8.78)$$

8.8.1 Horizontal direction

Since the right side does not depend on r , θ , or φ , $\frac{\nabla_h^2 Y}{Y}$ must not depend on θ , φ . Y_l^m which satisfies the boundary conditions on the spherical surface is known as a spherical harmonic function characterized by angular order l and azimuthal order m , and it satisfies

$$\nabla_h^2 Y_l^m = -l(l+1)Y_l^m. \quad (8.79)$$

8.8.2 Radial direction

Since the eigenfunction in the horizontal direction can be expressed with spherical harmonics, to find the eigenvalues and eigenfunctions for a homogeneous sphere we need to solve the eigenvalue problem of the ordinary differential equation

$$\frac{1}{r^2 R} \left(\frac{d}{dr} r^2 \frac{d}{dr} R \right) = \frac{l(l+1)}{r^2} - \frac{\omega^2}{c^2}. \quad (8.80)$$

The first term on the right hand side represents the horizontal wavenumber, and ω in the second term represents the eigenvalue. The eigenfunctions must satisfy the boundary conditions: (i) pressure $R = 0$ at the surface $r = r_0$, and (ii) finite and continuous at the center $r = 0$. The eigenfunctions are known as spherical Bessel functions $j_l(\omega/cr)$, $n_l(\omega/cr)$. Since j_l takes finite values at the origin and n_l diverges, we know from the second boundary condition that R is proportional to j_l . Furthermore, from the first boundary condition, it must satisfy $j_l(\omega/cr_0) = 0$, and from this condition we can determine the eigenfrequency ω .

§8.9 † Vector spherical harmonics

In the previous section, we considered a gas sphere, but to consider the oscillations of an elastic sphere, we need to consider vector value wave equations instead of scalar ones.

Normal modes of a spherically symmetric Earth can be classified into two types: spheroidal modes and toroidal modes^{note 11)}. All wavefields can be represented by a superposition of modes.

To represent spheroidal and toroidal modes, we need to extend spherical harmonics into vectors. The vector spherical harmonics are defined by

$$\mathbf{R}_{lm} = Y_l^m \hat{\mathbf{r}} \quad (8.81)$$

$$\mathbf{S}_{lm} = \nabla_h Y_l^m \quad (8.82)$$

$$\mathbf{T}_{lm} = -\hat{\mathbf{r}} \times \nabla_h Y_l^m \quad (8.83)$$

Because vector spherical harmonics are more complex, details are omitted, but for instance, eigenfunctions taking vector values satisfy relations with the horizontal Laplacian ∇_h^2 , such as $\nabla_h^2 \mathbf{T}_{lm} = -l(l+1) \mathbf{T}_{lm}$. $\mathbf{R}$ and $\mathbf{S}$ correspond to spheroidal modes, and $\mathbf{T}$ corresponds to toroidal modes. For details, refer to Dahlen and Tromp (1998) p. 872^(3)note 12).

8.9.1 Spheroidal mode

Spheroidal mode ${}_n S_l$: n represents the number of nodes in the radial direction, and l represents the number of nodes in the horizontal direction. P-SV waves can be represented by superpositions of spheroidal modes.

note 11) Here we assume horizontal heterogeneities are small enough to be ignored.

note 12) The definitions of spherical harmonics and vector spherical harmonics depend heavily on normalization, so it is important to first check which normalization is being followed.

${}_0S_0$ is a mode of uniform expansion and contraction. This mode can only be excited with observable amplitudes by giant earthquakes. Because it involves only volume changes, the attenuation due to shear deformation is small, and once excited, it is known to continue oscillating for several months.

${}_0S_1$ corresponds to translational motion and cannot exist due to physical constraints. ${}_1S_2$ can exist physically because the inner and outer parts move in opposite directions to conserve momentum (known as the Slichter mode), but its amplitude is small and it is very difficult to detect.

${}_0S_2$ is known as the football mode and is known to have the lowest eigenfrequency among normally observed modes, with a period of about 20 minutes. Figure 8.8 shows the spectrum of the ground motion record observed in Japan during the 2004 Sumatra earthquake. This figure shows that it is excited with an observable amplitude (2 mm) during giant earthquakes. The reason multiple peaks are visible is that the eigenperiod changes depending on the propagation direction due to the Coriolis force effect. Theoretically, Lord Kelvin calculated the eigenfrequencies in the 19th century by approximating the Earth as a homogeneous elastic sphere.⁽¹⁴⁾ Since each eigenfrequency reflects the internal structure of the Earth, active attempts at detection were made until the mid-20th century. The successful detection of the existence of Earth's free oscillations from observation records was first achieved during the 1960 Chilean earthquake using strainmeter records [Benioff et al. (1961)].⁽²⁾ Since then, observation examples of Earth's free oscillations have accumulated, and research to estimate the Earth's internal structure from the measured eigenfrequencies has been actively conducted.

The case of $n = 0$ is particularly called the fundamental mode and corresponds to Rayleigh waves.

8.9.2 Toroidal mode

Toroidal mode ${}_nT_l$: n represents the number of nodes in the radial direction, and l represents the number of nodes in the horizontal direction. SH waves can be described by superpositions of toroidal modes. The $n = 0$ case is particularly called the fundamental mode and corresponds to Love waves.

8.9.3 Correspondence between Earth's free oscillations and surface waves

In Figure 8.1 shown at the beginning of this chapter, we could observe wave groups of Rayleigh waves travelling around the Earth multiple times from the waveform recorded in Japan during the Chilean earthquake. Figure 8.10 shows its Fourier spectrum. The interval at which the wave groups appear corresponds to the time it takes for a surface wave to circumnavigate the Earth. Let us compare the spacing of spectral peaks in Figure 8.10 with the time required for a surface wave to circle the Earth. From the Jeans relation, letting the Earth's radius be R_E , the wavelength λ can be written as

$$\lambda_l = \frac{2\pi R_E}{l + 1/2} \quad (8.84)$$

The eigenfrequency f_l corresponding to l is obtained with phase velocity c_p as

$$f_l = \frac{c_p}{\lambda_l} = \frac{(l + 1/2)c_p}{2\pi R_E} \quad (8.85)$$

Roughly assuming $c_p = 4.5$ km/s, we get $f_{l+1} - f_l \sim 0.1$ mHz, which approximately matches the observed mode spacing.

Focusing on the envelope of the peaks, there seem to be periodic mounds. Let us consider these mounds. The absolute amplitude of the mode with azimuthal quantum number l (assuming $m = 0$), letting the angular distance between the source and observation station be Θ , can be written as

$$|P_l^0(\cos \Theta)| \sim \sqrt{\frac{2}{l\pi \sin \Theta}} |\cos[(l + 1/2)\Theta - \pi/4]| \quad (8.86)$$

Fixing Θ and changing l , we can see that a mound forms approximately every π/Θ and this repeats. For the Chilean earthquake, the angular distance to SGN is 154 degrees. In this case, waves gather at the antipode (directly opposite the source), so it is easier to estimate from the angle from the antipode (180-154) degrees^{note 13}. In this case, the width of a mound is estimated to be 7 in l , matching observations. These phenomena occur simply because the nodes of the mode repeatedly appear.

§8.10 Background Free Oscillations of the Earth

^{note 13})Of course, one can estimate at $\Theta = 150^\circ$, but in that case care should be taken since l is an integer.

<https://www.eri.u-tokyo.ac.jp/people/knishida/eng/seismology.html>

8.10.1 Discovery of Earth's Background Free Oscillations

In the Sun, it is known that turbulence near the surface continuously excites acoustic waves with a period of about 5 minutes. The velocity field of the solar surface is accurately observed from the ground, and the acoustic velocity structure and angular momentum distribution inside the Sun have been investigated in detail from the eigenperiods of the observed acoustic waves (Helioseismology). Kobayashi (1996) considered that a mechanism similar to the excitation mechanism of the 5-minute solar oscillations might also be effective for the Earth, Mars, and Venus. He theoretically estimated the magnitude of atmospheric disturbances and concluded that atmospheric disturbances cause observable levels of oscillations.

Following this, in 1998, Nawa et al. examined the data from the superconducting gravimeter at Syowa Station in Antarctica and discovered a phenomenon in which the solid Earth continuously oscillates in the period band of several hundred seconds even during periods of quiet seismic activity.⁽⁷⁾ This was the discovery of Earth's background free oscillations. Because the superconducting gravimeter records in Antarctica contained unique noise associated with tide level fluctuations, the detection was not yet considered conclusive. However, following the detection using the Antarctic data, they were successively detected at observation stations around the world, such as by LaCoste gravimeters installed at IDA stations⁽¹²⁾ and STS-1Z seismometers installed at IRIS/GEOSCOPE stations,⁽⁵⁾ and their existence became conclusive. Currently, they have been detected at more than 100 observation stations.

Subsequent research has revealed that ocean waves are also deeply involved in the excitation. Research in the broad framework of the atmosphere-ocean-solid Earth coupled system is ongoing^{note 14)}.

note 14) For details, see the review articles by Nishida [2013, 2017].^{(8), (9)}

§8.11 Bibliography

- [1] G. Arfken and H. Weber. *Mathematical Methods for Physicists*. Elsevier Science, 2013.
- [2] H. Benioff, F. Press, and S. Smith. Excitation of the free oscillations of the earth by earthquakes. *J. Geophys. Res.*, 66:605–619, 1961.
- [3] F. Dahlen and J. Tromp. *Theoretical Global Seismology*. Princeton University Press, Princeton, 1998.
- [4] F. Gilbert. Excitation of normal modes of the earth by earthquake sources. *Geophys. J. R. Astron. Soc.*, 22:223, 1971.
- [5] N. Kobayashi and K. Nishida. Continuous excitation of planetary free oscillations by atmospheric disturbances. *Nature*, 395(September):357–360, 1998.
- [6] G. Laske and R. Widmer-Schmidrig. 1.04 - theory and observations: Normal mode and surface wave observations. In G. Schubert, editor, *Treatise on Geophysics (Second Edition)*, pages 117–167. Elsevier, Oxford, Jan. 2015.
- [7] K. Nawa et al. Incessant excitation of the Earth’s free oscillations. *Earth Planet. Space*, 50:3–8, 1998.
- [8] K. Nishida. Earth’s background free oscillations. *Annu. Rev. Earth Planet. Sci.*, 41(1):719–740, May 2013.
- [9] K. Nishida. Ambient seismic wave field. *Proc. Jpn. Acad. Ser. B Phys. Biol. Sci.*, 93(7):423–448, 2017.
- [10] M. Saito. *DISPER80; a subroutine package for calculation of seismic normal-mode solutions*, pages 293–319. Acad. Press. San Diego, 1988.
- [11] M. Saitō. *The theory of seismic wave propagation*. 2016.
- [12] N. Suda, K. Nawa, and Y. Fukao. Earth’s background free oscillations. *Science*, 279:2089–2091, 1998.
- [13] H. Takeuchi and M. Saito. Seismology: Surface waves and free oscillations. In B. Bolt, editor, *Methods in Computational Physics*, volume 11, pages 217–295. Academic Press, New York, 1972.
- [14] W. Thomson. On the rigidity of the earth. *Philosophical Transactions of the Royal Society of London*, 153:573–582, Jan. 1863.

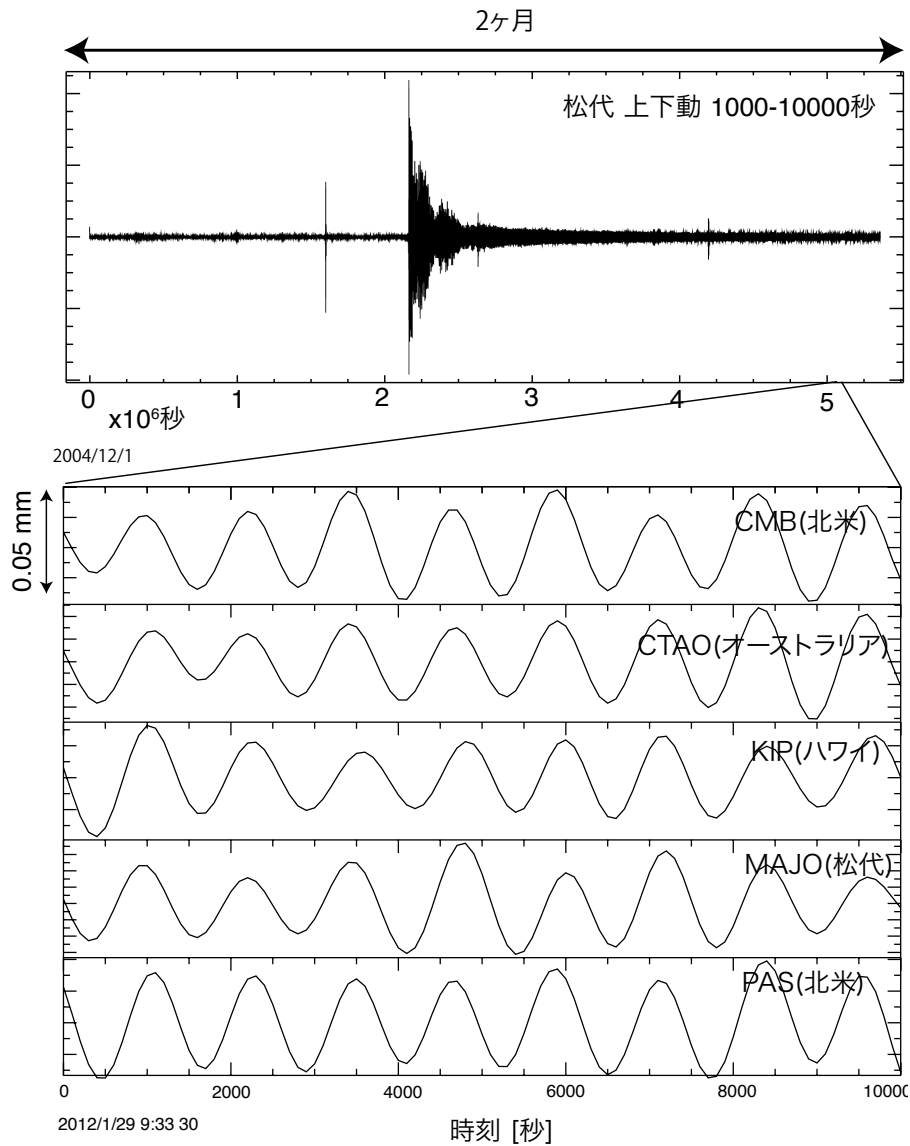


Figure 8.2: Ground motion records of the Sumatra earthquake. Filtered in the period band of 1000-10000 seconds. The upper panel shows the Earth oscillating for more than a month after the Sumatra earthquake. The lower panel is an enlarged view of the oscillations about one month after the earthquake. We can see simultaneous vertical movements around the world. This indicates that the entire Earth is repeatedly expanding and contracting.

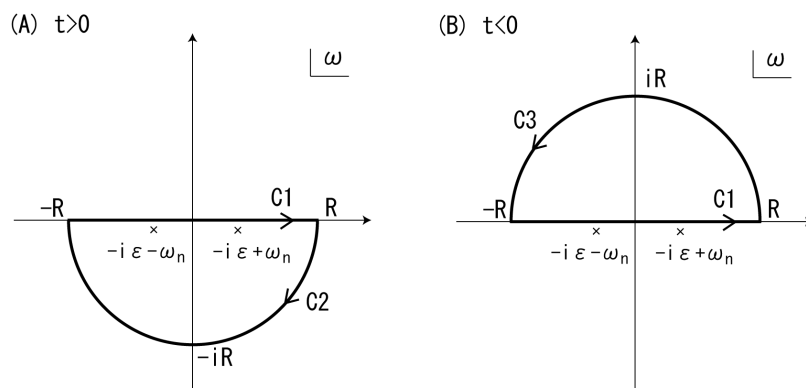


Figure 8.3: Integration path

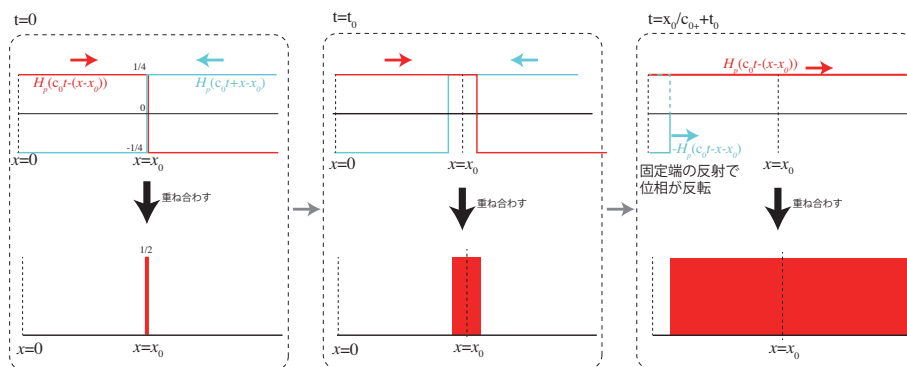


Figure 8.4: Schematic diagram.

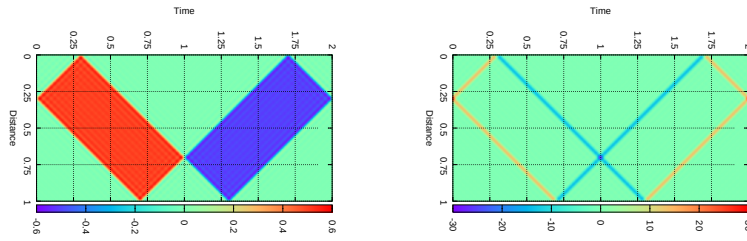


Figure 8.5: Calculated wave propagation.

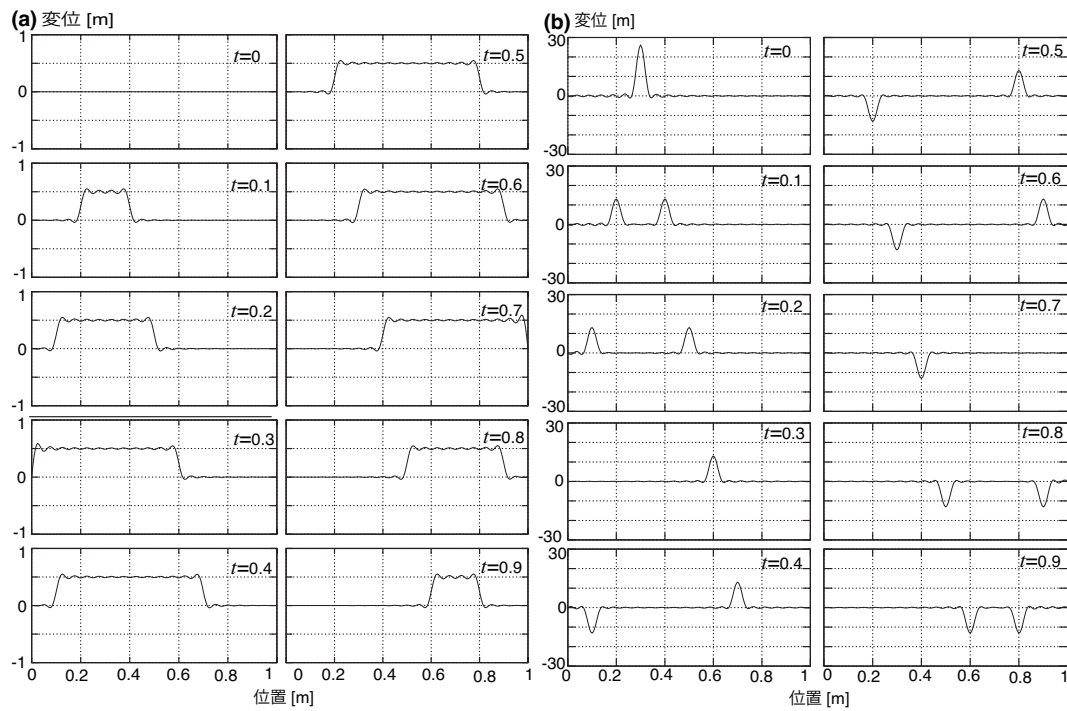


Figure 8.6: (a) Snapshot of calculated waveforms for the impulse response. (b) Snapshot of waveforms when the excitation term is substituted.

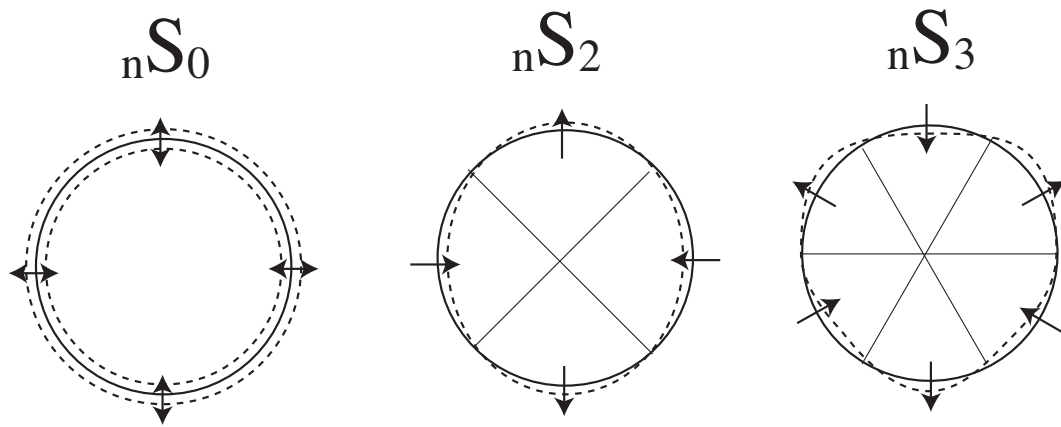


Figure 8.7: Examples of spheroidal modes.

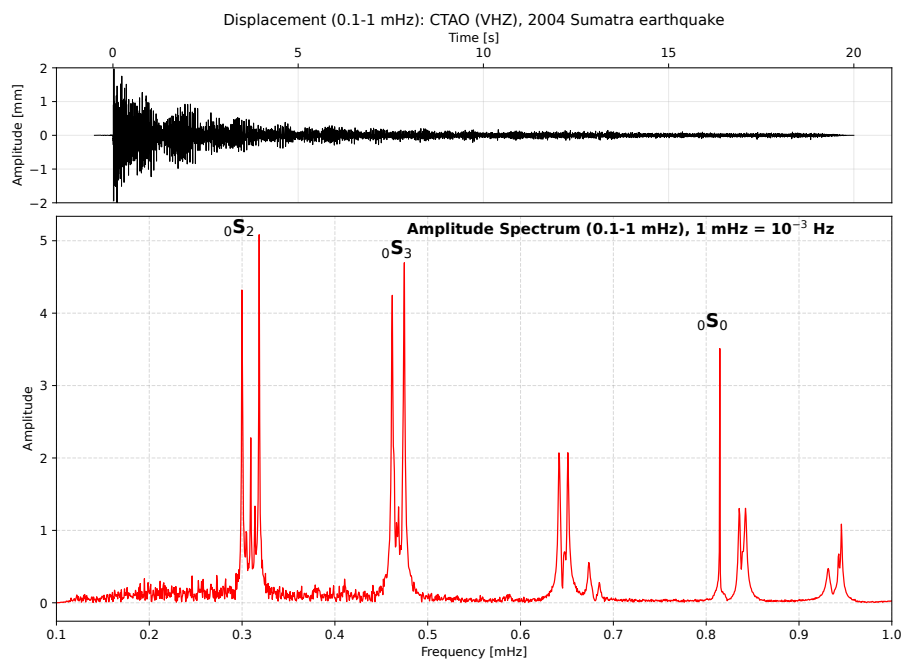


Figure 8.8: Ground motion record during the 2004 Sumatra earthquake.

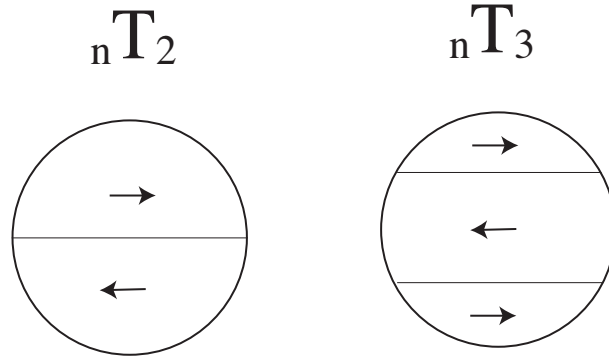


Figure 8.9: Examples of toroidal modes.

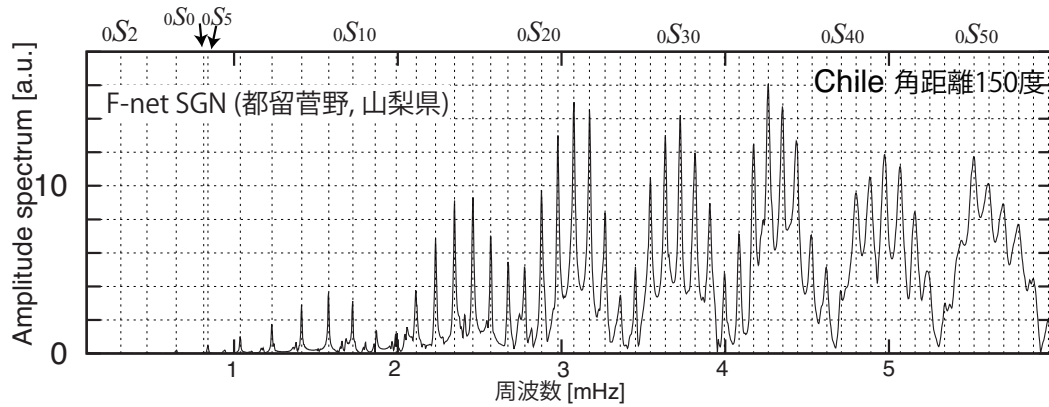


Figure 8.10: Fourier spectrum of the vertical ground motion recorded in Japan during the 2010 Chilean earthquake. We can observe many peaks corresponding to fundamental spheroidal modes. This is because the earthquake was shallow; for deep earthquakes, amplitude of overtones also becomes larger.